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Bachelor of Technology
in
COMPUTER SCIENCE AND ENGINEERING

Major Project Phase-II Report

AUTOMATED LEUKEMIA DETECTION BASED ON WHITE BLOOD CELL LOCALIZATION AND MULTITASK LEARNING

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ALL	Acute Lymphoblastic Leukaemia
AML	Acute Myeloid Leukaemia
BCCD	Blood Cell Count Dataset
CLL	Chronic Lymphocytic Leukaemia
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
CUDA	Compute Unified Device Architecture
DL	Deep Learning
F1	F1-Score (Harmonic Mean of Precision and Recall)
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
IoU	Intersection over Union
LIS	Laboratory Information System
mAP	Mean Average Precision
MLP	Multilayer Perceptron
NN	Neural Network
PNG	Portable Network Graphics
RBC	Red Blood Cell
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Network
SSD	Single Shot Detector
UI	User Interface
UNETR	UNet Transformer
ViT	Vision Transformer
WBC	White Blood Cell
WSI	Whole Slide Image
XAI	Explainable Artificial Intelligence

YOLO	You Only Look Once
ZIP	Compressed Archive Format

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ABSTRACT

Leukaemia is a life-threatening blood cancer caused by the excessive proliferation of abnormal white blood cells (blast cells) in the bone marrow and bloodstream. Early and accurate diagnosis is critical for effective treatment and improved survival rates. Conventional diagnosis relies on manual microscopic examination of blood smear images — a process that is time-consuming, error-prone, and difficult to scale.

This project presents an automated leukaemia detection system integrating deep learning-based object detection and classification into a unified pipeline. A YOLOv8-based detection module localizes red blood cells, white blood cells, and platelets with a mean average precision (mAP@0.50) of 0.986. Detected white blood cells are classified into five morphological subtypes using a Vision Transformer (ViT) model, achieving an overall accuracy of 0.99.

A multitask learning framework simultaneously predicts eleven fine-grained morphological attributes per cell, enabling transparent, attribute-driven blast cell identification aligned with clinical haematological criteria. The system computes blast percentage as the primary diagnostic parameter and applies a threshold-based decision rule for leukaemia indication. A Streamlit-based web interface supports single-image and batch processing, annotated outputs, and automated report generation, offering a clinically meaningful and interpretable solution for leukaemia screening and diagnostic support.

CHAPTER 1 INTRODUCTION

Leukemia is a type of blood cancer that affects the bone marrow and results in the excessive production of abnormal white blood cells, commonly known as blast cells. These abnormal cells interfere with normal blood cell production, leading to serious health conditions such as anemia, infections, and bleeding disorders. Early detection of leukemia is essential for effective treatment and improved survival rates.

Traditionally, leukemia diagnosis is performed through manual microscopic examination of peripheral blood smear images by trained hematologists. This process is time-consuming, requires significant expertise, and is prone to variability in interpretation. With the advancement of artificial intelligence, particularly deep learning, automated approaches have been developed to assist in medical image analysis and disease detection [1-2].

1.1 Project overview

The Leukemia Detection System is an AI-based diagnostic support tool designed to analyze blood smear images and identify abnormal white blood cells using deep learning techniques. The system integrates object detection and classification models to automate the identification of blast cells and calculate clinically relevant parameters as shown in the figure 1.

Deep learning models such as Convolutional Neural Networks (CNNs) have been widely used for image classification tasks, while advanced architectures like ResNet improve feature extraction performance [11]. In addition, object detection models such as YOLO enable real-time localization of white blood cells in complex images [9]. Recent developments in Vision Transformers (ViT) further enhance classification accuracy by capturing global image context [10].

The system combines these techniques to provide accurate detection, classification, and diagnostic support, thereby assisting medical professionals in decision-making.

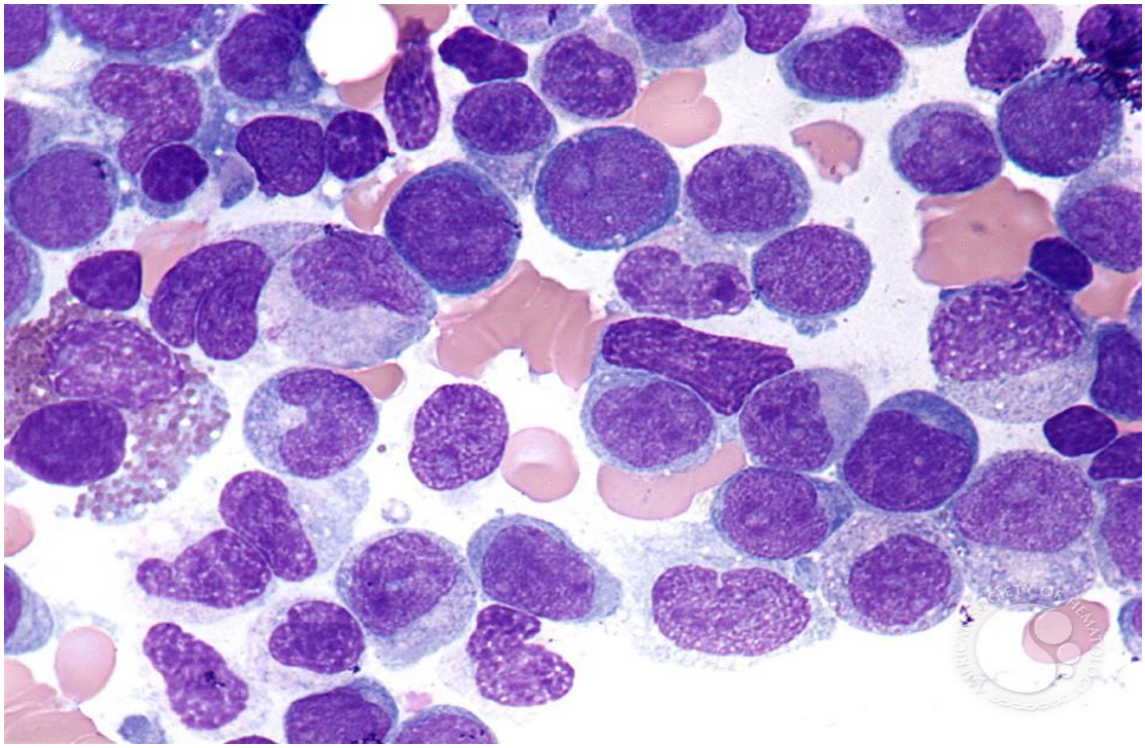


Figure 1: Sample peripheral blood smear image

1.2 Problem statement

The conventional method of leukemia diagnosis relies on manual analysis of blood smear images, which presents several challenges:

- Time-consuming and labor-intensive process
- Requires skilled hematologists for accurate diagnosis
- Susceptible to human error and inter-observer variability
- Difficult to scale for large populations

Although deep learning methods have been applied to medical image analysis, many existing systems either focus only on classification or lack efficient localization of cells [6], [7]. This creates a need for a more integrated and automated solution.

1.3 Objectives

The primary objectives of this project are as follows:

- To develop an automated blood smear image analysis system capable of detecting and localizing red blood cells, white blood cells, and platelets using a YOLO-based object detection model.
- To design a Vision Transformer-based classification module that accurately categorizes detected white blood cells into five morphological subtypes: neutrophils, eosinophils, monocytes, lymphocytes, and basophils.
- To implement a multitask learning framework that simultaneously predicts eleven fine-grained morphological attributes of each white blood cell, enabling interpretable and clinically meaningful cell characterization.
- To identify blast cells based on predicted morphological attribute profiles and compute quantitative hematological parameters including total WBC count, blast cell count, and blast percentage.
- To apply a threshold-based diagnostic decision rule that classifies a sample as leukemia-likely when the blast percentage meets or exceeds the clinical threshold of 50%.
- To evaluate the performance of the detection and classification modules using standard metrics including precision, recall, F1-score, and mean average precision (mAP).
- To develop a user-friendly, web-based diagnostic interface using Streamlit that supports single image and batch processing, interactive visualization, and automated report generation.

1.4 Scope of the project

The project focuses on the application of deep learning in analyzing peripheral blood smear images for leukemia detection.

The system includes:

- Detection of white blood cells using YOLO-based models [8], [16]
- Classification using deep learning architecture Vision Transformers [10]
- Computation of quantitative diagnostic parameters

The system is intended as a support tool and does not replace professional medical diagnosis.

1.5 Key Contributions and Novelty

The proposed system introduces several novel contributions that distinguish it from existing approaches in automated leukemia detection:

1.5.1. Unified End-to-End Detection and Classification Pipeline

Most existing systems treat white blood cell detection and classification as independent tasks. This project integrates both into a single cohesive pipeline where the output of the YOLO-based detection module directly feeds into the Vision Transformer classification module. This eliminates intermediate manual steps and reduces error propagation between stages.

1.5.2. Multitask Morphological Attribute Prediction

Rather than predicting only a single class label per cell, the proposed classification module simultaneously predicts eleven distinct morphological attributes per white blood cell using a multitask learning framework. These attributes — including nucleus shape, chromatin density, cytoplasm texture, granule type, and nuclear-to-cytoplasmic ratio — collectively characterize the cellular morphology in a manner consistent with clinical hematological assessment. With over 113,000 attribute labels from the WBC-ATT dataset, this fine-grained prediction provides a level of interpretability not present in any comparable automated system.

1.5.3. Attribute-Driven Blast Cell Identification

Existing blast detection systems typically train a binary classifier directly on blast vs. normal labels. The proposed system derives the blast/normal decision from the predicted morphological attribute profile, making the diagnostic reasoning transparent and explainable. This approach mirrors the actual diagnostic criteria used by hematologists and represents a clinically grounded methodology.

1.5.4. Multi-Class Blood Cell Quantification for Holistic Diagnosis

The system detects and counts not only white blood cells but also red blood cells and platelets. This enables the computation of a complete blood picture, including indicators of anemia (reduced RBC count) and thrombocytopenia (reduced platelet count), both of which are clinically associated with leukemia. No comparable automated pipeline provides this holistic quantification alongside blast prediction.

1.5.5. Clinically Interpretable Diagnostic Output

The system computes the blast percentage — the ratio of blast cells to total WBC count — as the primary diagnostic parameter, directly aligned with clinical diagnostic criteria. The threshold-based rule (blast percentage $\geq$ 50% indicating leukemia) mirrors established hematological practice, making the system's output directly usable as decision support in a clinical context.

1.5.6. Comprehensive Web-Based Diagnostic Interface

The Streamlit-based interface supports both single-image and batch processing, provides annotated visual outputs, interactive charts, morphological attribute breakdowns, and exports structured diagnostic reports. This level of clinical interface integration is absent from most research-level leukemia detection systems.

CHAPTER 2 PROBLEM DEFINITION

Leukemia is characterized by the abnormal proliferation of immature white blood cells in the bone marrow and bloodstream. These blast cells disrupt the normal production of blood components, leading to severe health complications. Accurate and early detection is critical for effective treatment.

Medical image analysis has gained importance with the advancement of deep learning techniques, which enable automated detection and classification of diseases from images [2]. However, applying these techniques effectively to leukemia detection remains a challenge.

2.1 Problem statement

The traditional approach to leukemia detection involves manual microscopic examination of blood smear images. This method has several limitations:

- High dependency on expert knowledge
- Time-consuming diagnostic process
- Possibility of inconsistent results
- Limited scalability for large-scale screening

Although deep learning approaches such as CNN-based classification have improved detection accuracy, many systems do not include proper localization of white blood cells [6], [7]. Similarly, some detection models lack precise classification capabilities.

2.2 Need for the proposed system

To overcome these challenges, an automated system is required that integrates both detection and classification.

Such a system should:

- Detect white blood cells efficiently using object detection models like YOLO [9]

- Classify cells accurately using Vision Transformers [10]
- Reduce manual effort and diagnostic time
- Provide consistent and reliable results

The integration of these techniques can significantly improve leukemia detection performance.

2.3 Proposed solution overview

The proposed system combines object detection and classification techniques into a unified framework.

- YOLO-based models are used for detecting white blood cells in images [8], [16]
- Classification is performed using deep learning models, including transformer-based approaches [10]
- Multitask learning concepts can further improve performance by handling multiple objectives simultaneously [20]

This approach enables accurate detection of blast cells and supports clinical diagnosis.

2.4 Significance of the problem

Early detection of leukemia is crucial for improving survival rates and treatment effectiveness. Automating the diagnostic process offers several benefits:

- Increased accuracy and consistency
- Faster analysis of medical images
- Reduced workload for medical professionals
- Potential for large-scale screening

Deep learning-based systems have the potential to transform medical diagnostics by providing efficient and reliable solutions [1], [2].

CHAPTER 3 LITERATURE REVIEW

There is a major advancement in areas such as, CNNs, object detection and transformers in automatic detection of leukemia from blood smear images [11-12]. CNN-based approaches work well on finely labeled datasets but require manual preprocessing and lack the ability to link blast cell detection with leukemia classification, making them less effective in practical applications [13-15]. YOLO-based detectors improve automation by directly detecting white blood cells, but they usually decouple the detection and classification tasks, causing possible errors to propagate across tasks [16-17]. Vision transformers are more effective in learning overall morphological characteristics but are usually restricted to single-task classification tasks [18-20]. A comparative study of recent blood smear image analysis methods along with highlighting their key contributions and limitations are listed in Table 1.

Table 1: Comparative Table of Recent Leukemia Detection Techniques

Ref. No.	Reference	Method Used	Key Contribution	Limitation
[1]	LeCun et al. (2015)	General Deep Learning (CNN/RNN/MLP)	Established the fundamental framework for hierarchical feature learning that replaces hand-crafted features	High dependence on massive labeled datasets and high computational cost for training
[2]	Litjens et al. (2017)	Survey of DL in Medical Imaging	Systematically categorized AI applications in pathology, including cell classification and segmentation	Highlighted that most models at the time lacked clinical validation on "real-world" noisy data
[3]	Acevedo et al. (2020)	Morphological Attribute Annotation (WBC-ATT)	Provided a densely annotated dataset with 11 morphological attributes (nucleus, granules, etc.) for Explainable AI (XAI)	Focused primarily on normal cell categories; lacks specific labels for all leukemia subtypes
[4]	Tan et al. (2022)	TBXPCB Multi-focus Dataset	Introduced a comprehensive dataset using multi-focus image	The multi-focus approach requires more storage and

			stacks to improve classification accuracy	slower image acquisition hardware
[5]	Abbas et al. (2022)	GAN + CNN Transfer Learning	Used GANs to generate synthetic images, overcoming data imbalance	Synthetic images may lack the fine textures needed for clinical accuracy
[6]	Rezatofighi et al. (2019)	DRFA-Net (Attention)	Automated WBC localization to focus features on relevant cell regions	Inaccurate if initial detection fails in dense or overlapping samples
[7]	Ramesh et al. (2020)	CNN-based leukemia detection	Demonstrated CNN applicability for automated leukemia detection from blood smears	Limited to binary classification; no cell localization or attribute analysis
[8]	Shakarami et al. (2021)	YOLO-based WBC detection	Applied YOLO models specifically for automated WBC detection and classification	Detection and classification are decoupled; no blast-specific analysis
[9]	Redmon et al. (2016)	Original YOLO architecture	Introduced real-time single-pass object detection framework	Early version limited in detecting small or overlapping objects
[10]	Dosovitskiy et al. (2021)	Vision Transformer (ViT)	Demonstrated transformer-based global attention for image classification	Restricted to single-task classification; requires large datasets for training
[11]	He et al. (2016)	ResNet deep residual learning	Introduced skip connections enabling very deep CNNs without degradation	Local receptive field limits global morphological feature capture
[16]	Jocher et al. (2023)	YOLOv8	Improved detection accuracy and speed with anchor-free architecture	No integration with downstream classification or diagnostic reporting
[18]	Hatamizadeh et al. (2022)	UNETR (Transformer for segmentation)	Extended transformers to 3D medical image segmentation	Focused on segmentation; not applicable to cell-level classification
[20]	Caruana (1997)	Multitask Learning framework	Established theoretical foundation for simultaneous multi-objective learning	Theoretical work; not applied to medical image analysis

3.1 Limitations of existing methods

Despite the progress made in this field, several challenges remain:

- Many models rely solely on classification without accurate localization of cells
- CNN-based approaches may fail to capture global dependencies in images
- Manual preprocessing steps can affect consistency
- Some systems lack integration of detection and classification in a unified pipeline

These limitations indicate the need for more robust and integrated approaches.

3.2 Motivation for proposed work

Based on the reviewed literature, there is a clear need for a system that combines efficient detection and accurate classification within a single framework. The integration of YOLO for localization and Vision Transformers for classification provides a balanced approach that leverages both speed and accuracy.

The proposed work is motivated by the need to improve diagnostic reliability, reduce manual effort, and provide an automated solution for leukemia detection.

3.2 Gap Analysis of Existing Methods

To clearly establish the novelty of the proposed system, Table 2 presents a systematic comparison of key capabilities across representative existing works and the proposed system.

Table 2 Capability Comparison of Existing Methods and the Proposed System

Capability	**CNN-based Methods [6,7,13]**	**YOLO-based Methods [8,16]**	**ViT-based Methods [10,18]**	**Proposed System**
WBC Localization	Partial / None	Yes	No	Yes
WBC Subtype Classification	Yes	No	Yes	Yes
Morphological Attribute Prediction	No	No	No	Yes (11 attributes)
Blast Cell Identification	Binary only	No	No	Attribute-driven
RBC / Platelet Counting	No		No	Yes
Blast Percentage Computation	No	No	No	Yes
Multitask Learning	No	No	No	Yes
Diagnostic Report Generation	No	No	No	Yes
Clinical Interpretability	Low	Low	Medium	High

CHAPTER 4 PROJECT DESCRIPTION

This chapter describes the overall design and working of the proposed leukemia detection system. The system integrates deep learning techniques for automated analysis of blood smear images. It combines object detection and classification models to identify white blood cells and detect blast cells for diagnostic support.

The system is designed to improve accuracy and efficiency by reducing manual intervention and utilizing advanced architectures such as YOLO for detection and Vision Transformers for classification [9], [10].

4.1 System architecture

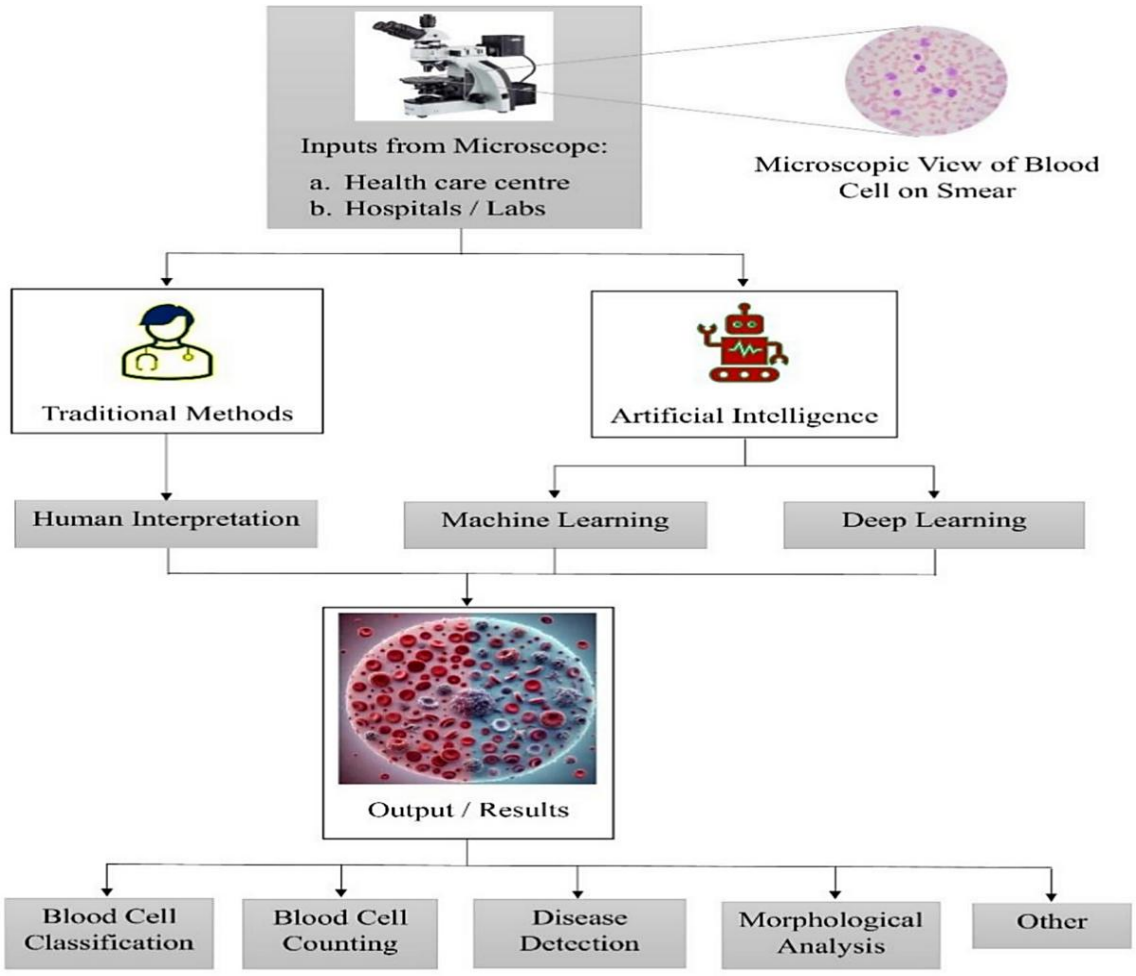


Figure 2:Proposed system architecture

Figure 2 shows the architecture of the proposed system consists of multiple interconnected modules that process the input image step by step.

The major components of the system include:

- Input module: Accepts blood smear images from the user
- Preprocessing module: Enhances image quality and prepares it for analysis
- Detection module: Identifies and localizes white blood cells using YOLO-based models [9], [16]
- Classification module: Classifies detected cells into blast and normal categories using Vision Transformers [10]
- Output module: Generates annotated images and diagnostic reports

This architecture ensures efficient data flow and accurate detection of leukemia-related features.

4.2 Design Rationale and Architecture Decisions

The selection of each architectural component in the proposed system is motivated by both technical performance considerations and the specific requirements of clinical blood smear analysis.

4.2.1 Choice of YOLO for Detection

Object detection architectures such as Faster R-CNN and SSD have been applied to medical image analysis, but their two-stage detection approach introduces latency unsuitable for near-real-time clinical use. YOLO performs detection in a single forward pass, achieving a favorable balance between speed and accuracy. YOLOv8, the version employed in this system, introduces an anchor-free detection head that improves localization of small objects such as platelets, which are a known challenge in blood smear analysis. The model is trained on the TXL-PBC dataset, which aggregates annotations from multiple sources to ensure robust generalization across varying image conditions.

4.2.2 Choice of Vision Transformer for Classification

Convolutional Neural Networks extract features through local receptive fields, which limits their ability to capture global spatial relationships within a cell. White blood cell morphology, however, involves global characteristics — the relationship between nucleus shape, cytoplasm extent, and granule distribution across the entire cell. Vision Transformers address this through self-attention mechanisms that relate all image patches simultaneously, making them inherently better suited for capturing the holistic morphological features required for accurate WBC classification and attribute prediction.

4.2.3 Choice of Multitask Learning for Attribute Prediction

Training eleven separate classifiers for eleven morphological attributes would be computationally expensive and would fail to exploit the correlations between attributes. For example, blast cells tend to simultaneously exhibit irregular nucleus shapes, high nuclear-to-cytoplasmic ratios, and absent granules. Multitask learning leverages a shared ViT encoder whose representations benefit all attribute prediction heads simultaneously, improving generalization and computational efficiency. The shared representation also implicitly learns correlations between attributes, producing more coherent and clinically meaningful predictions.

4.3 Module description

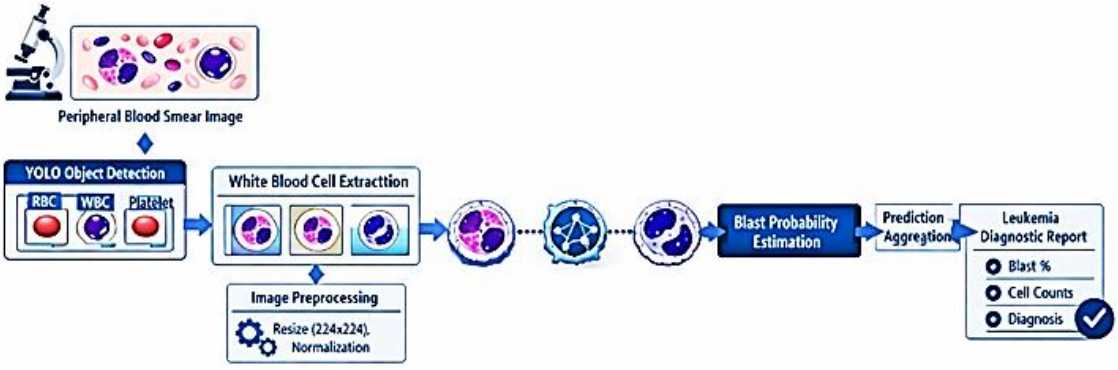


Figure 3: Complete end-to-end workflow for automated blast prediction and diagnostic report generation

As per the Figure 3 the complete workflow description is discussed below.

4.3.1 Image preprocessing

The preprocessing module prepares the input image for analysis by performing operations such as resizing, normalization, and noise reduction. These steps improve image quality and ensure consistency in model input.

4.3.2 Detection module

The detection module uses YOLO-based models to identify and localize white blood cells in the input image. YOLO is a real-time object detection algorithm known for its speed and accuracy [9]. Advanced versions such as YOLOv8 further improve detection performance [16].

4.3.3 Classification module

The classification module categorizes detected cells into blast and normal cells. Deep learning model Vision Transformers are used for this task. Vision Transformers provide improved performance by capturing global relationships in the image [10].

4.3.4 Report generation module

The report generation module computes important hematological parameters such as total white blood cell count, blast cell count, and blast percentage. Based on these values, the system provides a diagnostic interpretation.

CHAPTER 5 REQUIREMENTS

This chapter describes the functional and non-functional requirements of the proposed leukemia detection system. The requirements define the expected behavior of the system and the constraints under which it operates. The system is designed to analyze blood smear images using deep learning techniques and provide diagnostic support for leukemia detection.

5.1 Functional requirements

Functional requirements specify the core operations that the system must perform.

5.1.1 Image input

- The system must allow users to upload blood smear images
- The input should support standard image formats such as JPG and PNG
- The system should validate the uploaded image before processing

5.1.2 Image preprocessing

- The system must preprocess the input image to enhance quality
- Operations include resizing, normalization, and noise reduction
- Preprocessing ensures compatibility with deep learning models

5.1.3 White blood cell detection

- The system must detect and localize white blood cells in the image
- Detection should be performed using YOLO-based object detection models [9], [16]
- The system must generate bounding boxes around detected cells

5.1.4 Cell classification

- The system must classify detected cells into:

  - Blast cells
  - Normal cells
- Classification should be performed using Vision Transformers [10-11]

5.1.5 Quantitative analysis

- The system must calculate:
  - Total white blood cell count
  - Number of blast cells
  - Number of normal cells
  - Blast percentage
- Blast percentage is used as a key diagnostic parameter

5.1.6 Diagnostic decision support

- The system should provide a diagnostic interpretation based on blast percentage
- The system must follow a threshold-based rule for leukemia indication

5.1.7 Report generation

- The system must generate a structured diagnostic report
- The report should include:
  - Annotated image
  - Cell counts
  - Blast percentage
  - Diagnostic result

5.1.8 User interface

- The system must provide a simple and user-friendly interface
- Users should be able to upload images and view results easily

5.2 Non-functional requirements

Non-functional requirements define system quality attributes.

5.2.1 Performance

- The system should process images efficiently with minimal delay
- Detection and classification should be performed in near real-time using optimized models [9]

5.2.2 Accuracy

- The system should provide high accuracy in detection and classification
- Deep learning models must be trained effectively to ensure reliable results [1-2]

5.2.3 Reliability

- The system should produce consistent results for similar inputs
- It should handle errors gracefully

5.2.4 Scalability

- The system should be capable of handling large datasets
- It should support future enhancements and integration

5.2.5 Usability

- The interface should be easy to use with minimal training
- Results should be presented in a clear and understandable format

5.2.6 Maintainability

- The system should be modular for easy updates and improvements
- Code structure should allow integration of new models

5.3 Software requirements

The system requires the following software:

- Operating System: Windows/Linux
- Programming Language: Python
- Libraries:
    - Deep learning frameworks (TensorFlow/PyTorch)
    - OpenCV for image processing
- Tools:
    - Streamlit for user interface

CHAPTER 6 METHODOLOGY

This chapter describes the methodological approach followed to develop the leukemia detection system. The methodology focuses on the step-by-step processing of blood smear images using deep learning techniques, including detection, classification, blast prediction, and quantitative analysis. The approach integrates multiple models into a unified pipeline to achieve accurate and efficient diagnosis.

6.1 Overall methodology

The proposed methodology is designed as a sequential pipeline consisting of multiple stages:

1. Image acquisition
2. Image preprocessing
3. Blood cell detection
4. WBC classification
5. Blast prediction
6. Quantitative analysis
7. Diagnostic decision

Each stage contributes to transforming raw input images into meaningful diagnostic outputs.

6.2 Image acquisition

The system accepts peripheral blood smear images as input. These images are obtained from microscopic examination and may vary in resolution, lighting conditions, and quality. The system standardizes these inputs for consistent processing.

6.3 Image preprocessing

Before analysis, the input image undergoes preprocessing to improve quality and consistency. The preprocessing steps include:

- **Resizing:** Standardizes image dimensions
- **Normalization:** Scales pixel values
- **Noise reduction:** Removes unwanted artifacts

These steps improve model performance and detection accuracy.

6.4 Blood cell detection

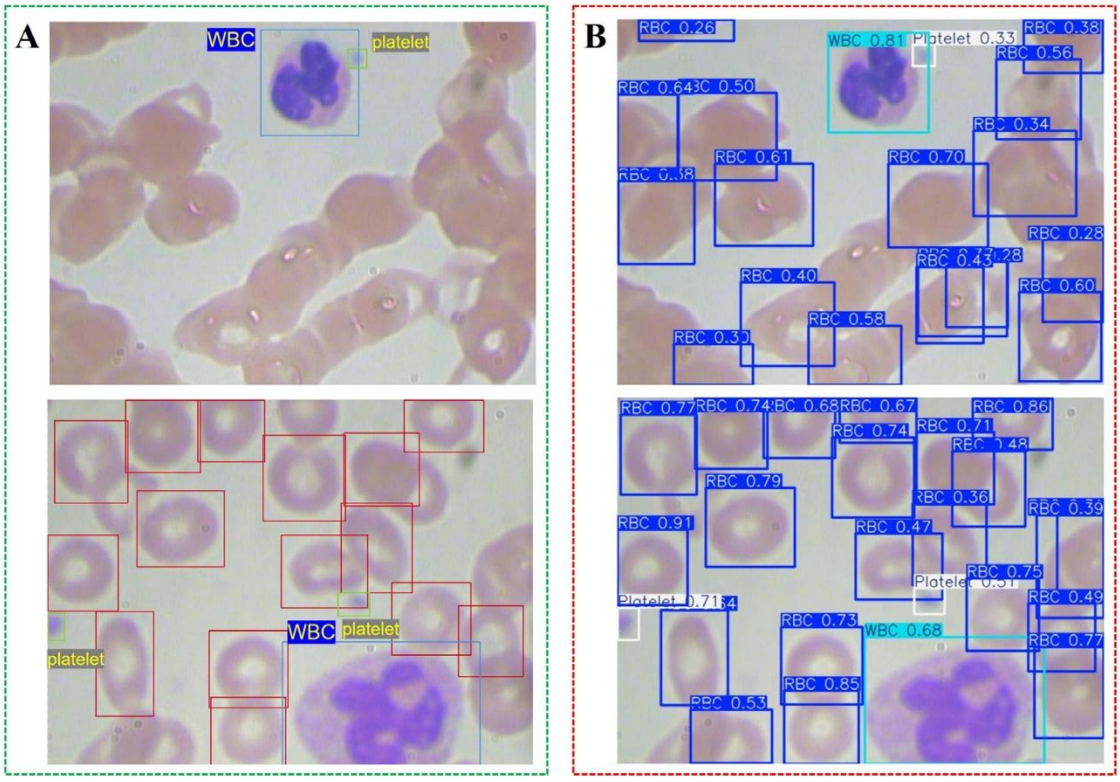


Figure 4 : Detection of blood cells

In this stage, as shown in figure 4 the system detects and localizes different types of blood cells in the image. The detection is performed using YOLO-based object detection models [9], [16].

This stage identifies:

- Red blood cells (RBCs)
- White blood cells (WBCs)

- Platelets

The RBC, WBC, and platelet counts are needed to identify abnormal blood cell levels caused by bone marrow dysfunction, which helps support leukemia diagnosis.

- RBC count: Usually decreases (anemia) due to reduced normal red blood cell production in the bone marrow.
- WBC count: Often increases abnormally with the presence of immature blast cells**.**
- Platelet count: Typically decreases (thrombocytopenia) because platelet production is suppressed.

6.5 WBC classification

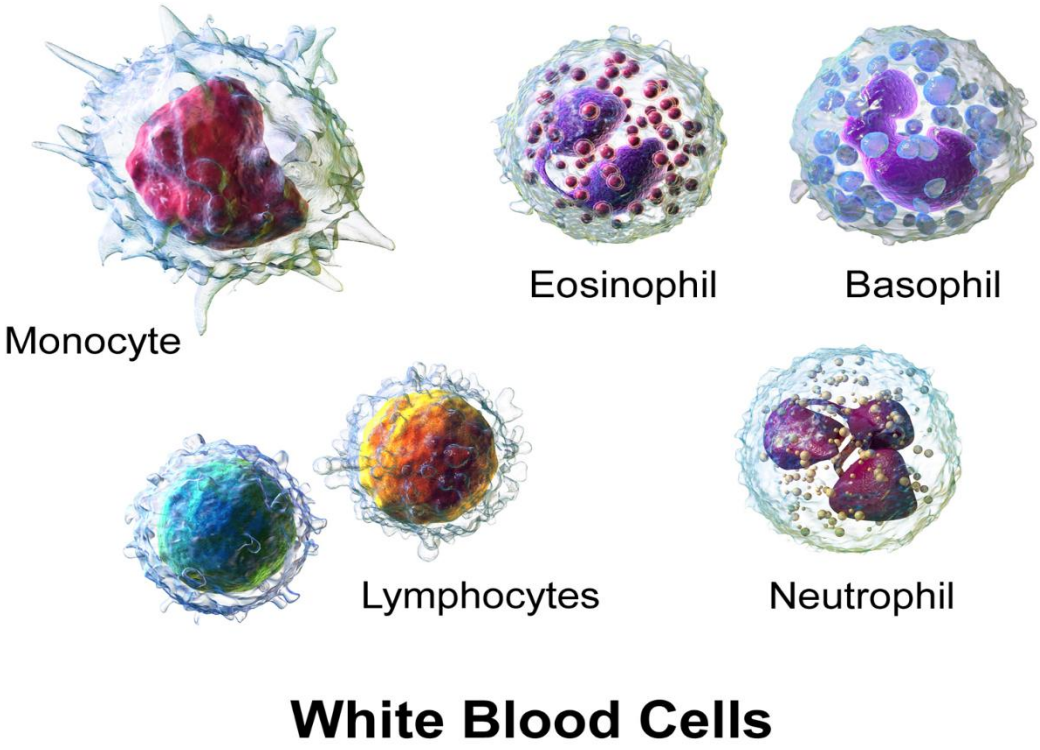


Figure 5: Classification of white blood cells

After detection, only the white blood cells are extracted and passed to the classification module. This stage classifies WBCs into different categories based on their morphological features as shown is figure 5.

Vision Transformers are used for this purpose, enabling accurate feature extraction and classification [10-11].

6.6 Blast prediction

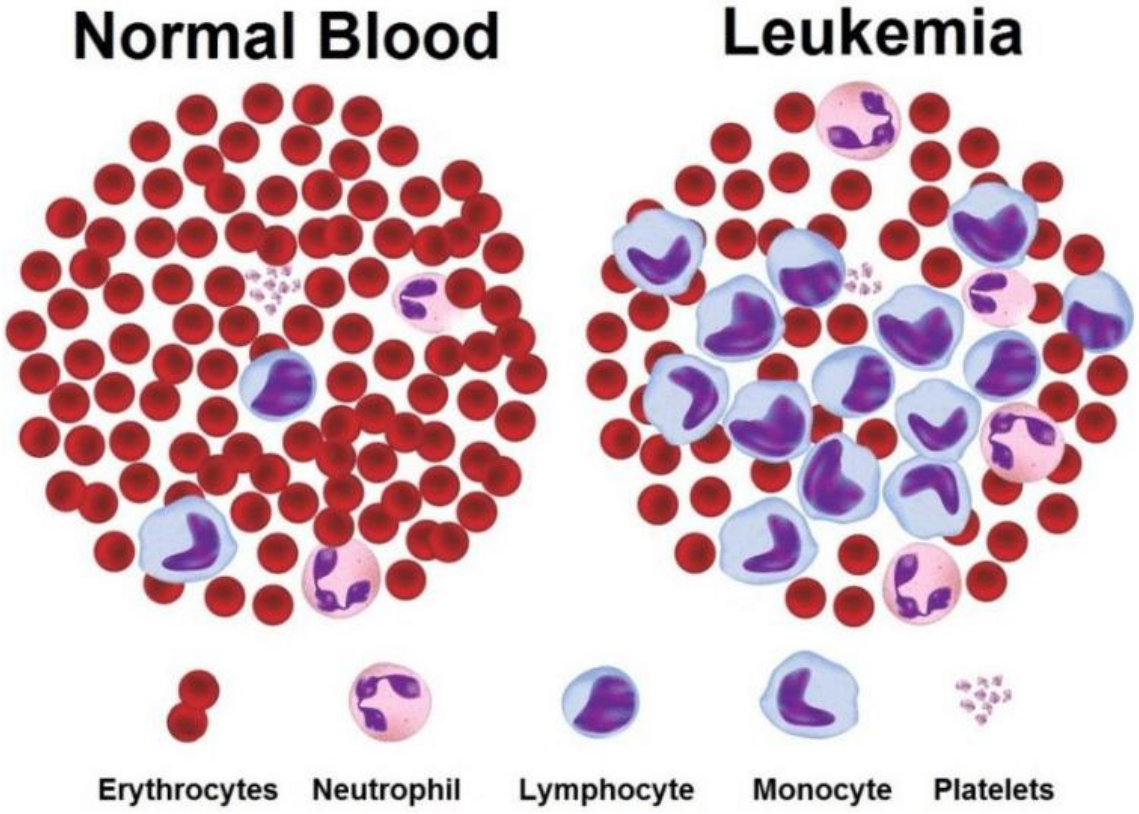


Figure 6: Blast cell prediction

In this stage, as shown in figure 6 the system identifies whether a classified white blood cell is a blast cell or a normal cell. This is a critical step in leukemia detection.

The model analyzes features such as:

- Cell Size (2 classes): big, small
- Cell Shape (2 classes): irregular, round

- Nucleus Shape (6 classes): irregular, segmented-bilobed, segmented-multilobed, unsegmented-band, unsegmented-indented, unsegmented-round
- Nuclear-to-Cytoplasmic Ratio (N:C Ratio) (2 classes): high, low
- Chromatin Density (2 classes): densely, loosely
- Cytoplasm Vacuoles (2 classes): present, absent
- Cytoplasm Texture (2 classes): clear, frosted
- Cytoplasm Color (3 classes): blue, light blue, purple blue
- Granule Type (4 classes): coarse, absent, round, small
- Granule Color (4 classes): absent, pink, purple, red
- Granularity (2 classes): present, absent This stage directly contributes to the diagnostic decision-making process.

6.6.1 Multitask Vision Transformer Architecture for Attribute Prediction

The classification and attribute prediction module is implemented as a multitask Vision Transformer. The architecture consists of the following components:

6.6.1.1 Shared Encoder:

The input cell image — cropped from the detected bounding box — is divided into fixed-size patches (16×16 pixels), linearly embedded, and processed through a standard ViT encoder consisting of multiple transformer blocks with multi-head self-attention and feed-forward layers. This shared encoder produces a unified feature representation that captures global morphological information from the entire cell image.

6.6.1.2 Task-Specific Classification Heads:

The [CLS] token output of the shared encoder is passed to eleven independent fully connected classification heads, one per morphological attribute. Each head produces a probability distribution over the classes defined for that attribute. For example, the nucleus shape head outputs probabilities over six classes (irregular, segmented-bilobed,

segmented-multilobed, unsegmented-band, unsegmented-indented, unsegmented-round), while the cell size head outputs probabilities over two classes (big, small).

6.6.1.3 Loss Function:

The total training loss is the unweighted sum of cross-entropy losses from all eleven task heads:

$$L_{Total} = \sum_{i=1}^{11} L_{CE}^{(i)} \qquad \text{Eq 1}$$

where L_CE(i) is the cross-entropy loss for attribute i. Each head is trained simultaneously through backpropagation of the combined loss, ensuring that the shared encoder learns representations useful across all attribute prediction tasks.

6.6.1.4 Blast/Normal Decision:

Following attribute prediction, the predicted attribute profile of each WBC is evaluated against morphological patterns characteristic of blast cells. Cells exhibiting attributes consistent with blast morphology — such as irregular nucleus shape, high nuclear-to-cytoplasmic ratio, loosely packed chromatin, and absent granules — are classified as blast cells. The blast percentage is subsequently computed from the ratio of blast-classified cells to the total WBC count.

6.7 Quantitative analysis

After blast prediction, the system computes important hematological parameters:

- Total white blood cell count
- Number of blast cells
- Number of normal cells
- Blast percentage

The blast percentage is calculated as:

$$\text{Blast Percentage} = \left(\frac{\text{Number of Blast Cells}}{\text{Total WBC Count}}\right) \times 100 \qquad \text{Eq 2}$$

This parameter is essential for leukemia diagnosis.

6.8 Diagnostic decision rule

The system applies a threshold-based rule:

- If blast percentage ≥ 50% → Leukemia likely
- If blast percentage < 50% → Non-leukemic

This rule provides decision support based on computed results.

6.9 Integration of models

The methodology integrates detection, classification, and prediction models into a unified pipeline. The detection output is used for classification, and classification results are used for blast prediction.

Multitask learning concepts can be applied to improve efficiency and performance by optimizing multiple objectives simultaneously [20].

CHAPTER 7 EXPERIMENTATION

This chapter presents the experimental setup used to evaluate the proposed leukemia detection system. It includes details of the datasets used, preprocessing techniques, model training configuration, testing procedures, and evaluation metrics. The objective of this experimentation is to validate the effectiveness of the system in detecting blood cells, classifying white blood cells, and predicting blast cells accurately.

7.1 Dataset description

The proposed system utilizes two primary datasets for detection and classification tasks.

7.1.1 TXL-PBC dataset

The TXL-PBC dataset is designed to address the limitations of existing public blood cell datasets, such as limited availability and inconsistent annotations. It combines data from multiple sources to provide a comprehensive dataset for detection.

The dataset integrates images from the following sources:

- BCCD: Approximately 12,500 annotated blood smear images for RBC, WBC, and platelet detection
- BCDD: Peripheral blood smear dataset for blood cell localization
- PBC: Labeled peripheral blood smear images for WBC subtype classification
- Raabin-WBC: Large-scale dataset (~17,000 images) with detailed WBC subtype annotations

Dataset characteristics:

- Total images: 1,260 high-quality microscopic images
- Total annotations: 18,143 bounding boxes covering RBCs, WBCs, and platelets

This dataset is used in the detection stage to identify and localize blood cells.

7.1.2 WBC-ATT dataset

The WBC-ATT dataset is used for detailed white blood cell classification and attribute-based analysis.

Dataset characteristics:

- Total images: 10,298

Class distribution:

- Neutrophils: 3,329 images
- Eosinophils: 3,117 images
- Monocytes: 1,420 images
- Lymphocytes: 1,214 images
- Basophils: 1,218 images

Each image is annotated with detailed morphological attributes.

Annotated attributes include:

- Cell size: big, small
- Cell shape: irregular, round
- Nucleus shape: irregular, segmented-bilobed, segmented-multilobed, unsegmented-band, unsegmented-indented, unsegmented-round
- Nuclear-to-cytoplasmic ratio: high, low
- Chromatin density: densely, loosely
- Cytoplasm vacuoles: present, absent
- Cytoplasm texture: clear, frosted
- Cytoplasm color: blue, light blue, purple blue
- Granule type: coarse, absent, round, small
- Granule color: absent, pink, purple, red
- Granularity: present, absent

This results in more than 113,000 attribute labels, which are used to train the interpretable attribute branch of the Vision Transformer model.

7.2 Dataset Splits

The datasets are divided into training, validation, and test subsets to ensure fair evaluation and prevent overfitting.

For the TXL-PBC detection dataset:

- Training set: 70% of images (approximately 882 images)
- Validation set: 15% of images (approximately 189 images)
- Test set: 15% of images (approximately 189 images)

For the WBC-ATT classification dataset:

- Training set: 70% of images (approximately 7,209 images)
- Validation set: 15% of images (approximately 1,545 images)
- Test set: 15% of images (approximately 1,544 images)

Stratified splitting is applied to ensure proportional class representation across all subsets.

7.3 Data Augmentation

To improve model generalization and address class imbalance, the following augmentation techniques are applied during training:

For the detection model (TXL-PBC):

- Random horizontal and vertical flipping
- Random rotation (±15 degrees)
- Mosaic augmentation (combining four images into one training sample)
- Random scaling and translation
- Color jitter (brightness, contrast, saturation adjustment)

For the classification model (WBC-ATT):

- Random horizontal flipping
- Random rotation (±20 degrees)
- Color jitter to simulate staining variation across laboratories
- Random cropping and resizing
- Gaussian noise addition to simulate imaging artifacts

Stain normalization is applied as a preprocessing step to reduce variability introduced by different staining protocols across source datasets.

7.4 Hyperparameters and Training Configuration

Table 3 Training Hyperparameters for Detection and Classification Models

Parameter	**Detection Model (YOLO)**	**Classification Model (ViT)**
Input Image Size	640 × 640 px	224 × 224 px
Batch Size	16	32
Learning Rate	0.01 (cosine decay)	0.0001
Optimizer	SGD with Momentum	Adam
Epochs	100	50
Early Stopping Patience	10 epochs	10 epochs
Pretrained Weights	COCO Dataset	ImageNet-21k

7.5 Image preprocessing

Before training, all images undergo preprocessing to ensure consistency and improve model performance.

For the TXL-PBC dataset:

- Full blood smear images are resized to a standard input size
- Images are normalized to reduce intensity variations
- Processed images are passed to the YOLO-based detection model

- Detected white blood cells are cropped using predicted bounding boxes
- For the WBC-ATT dataset:
- Cropped cell images are resized according to Vision Transformer input requirements
- Pixel intensities are normalized to reduce variations caused by staining and lighting

These preprocessing steps standardize the data and enhance model accuracy.

7.6 Training setup

The models are trained using deep learning frameworks with appropriate configurations.

- Detection model: YOLO-based architecture for blood cell detection [9], [16]
- Classification model: Vision Transformer and CNN-based models for WBC classification [10], [11]
- Optimizer: Adam optimizer for efficient learning [12]

Training involves adjusting parameters such as:

- Learning rate
- Batch size
- Number of epochs

The models learn features and patterns associated with different blood cell types and blast cells.

7.7 Testing procedure

After training, the system is evaluated using unseen test data.

- Input images are processed through the detection model

- Detected cells are classified into WBC types
- Blast prediction is performed
- Results are compared with ground truth

This ensures the system performs effectively on new data.

7.8 Evaluation metrics

The performance of the system is evaluated using standard metrics:

- Accuracy
- Precision
- Recall
- F1-score

For detection:

- Mean Average Precision (mAP) 0.50 & 0.95

These metrics provide a comprehensive evaluation of system performance.

7.9 Experimental results overview

The experimental results indicate that:

- The detection model effectively identifies blood cells
- The classification model accurately categorizes white blood cells
- The blast prediction module reliably identifies abnormal cells

The integrated approach improves overall system performance.

CHAPTER 8 RESULTS AND DISCUSSION

This chapter presents the results obtained from the implementation of the leukemia detection system. The results include outputs from blood cell detection, white blood cell classification, blast prediction, and quantitative analysis. The performance of the system is evaluated based on the observed outputs and computed metrics.

8.1 Blood cell detection results

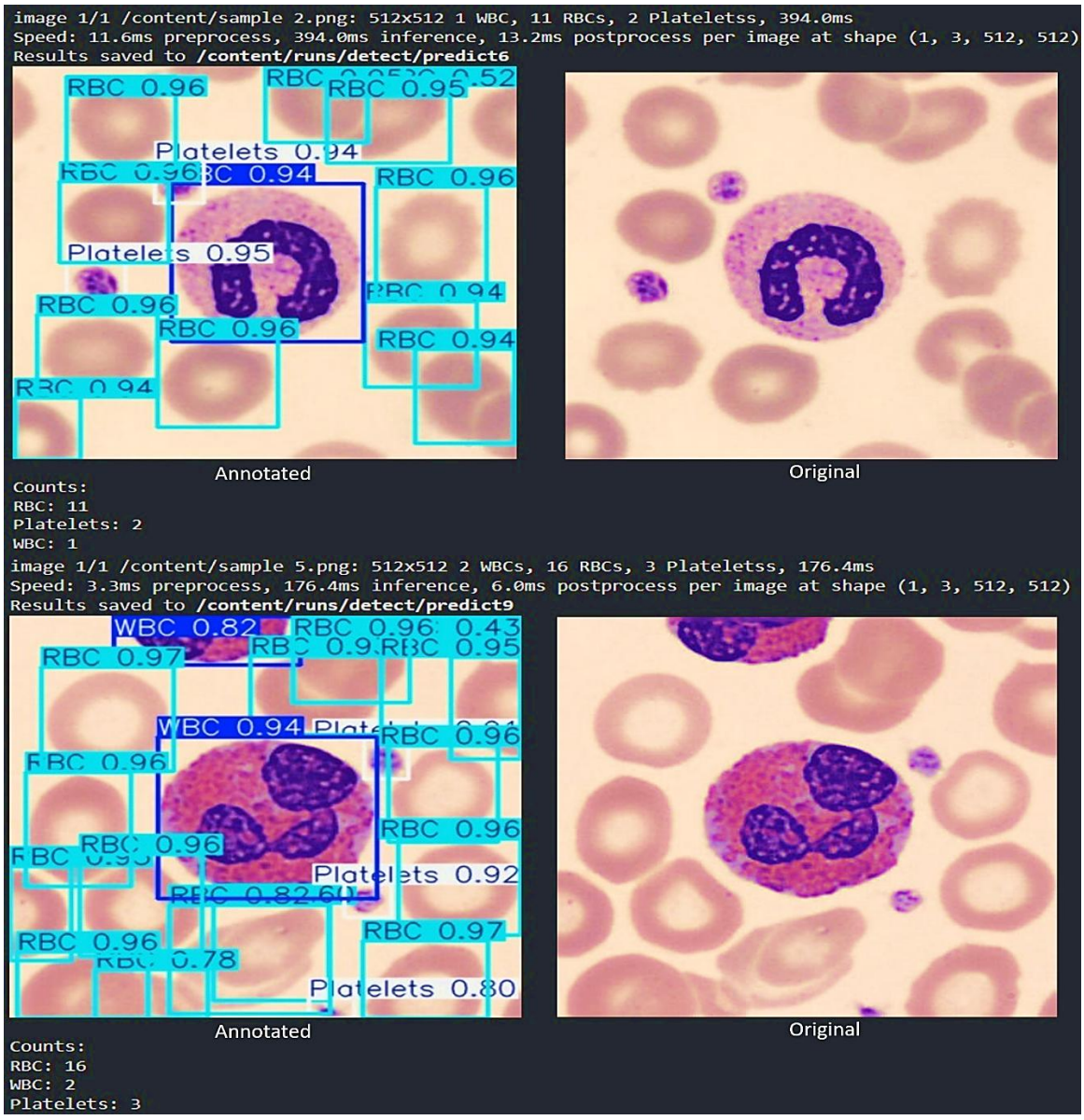


Figure 7: Blood cell detection output

The detection model successfully identifies and localizes different types of blood cells, including red blood cells, white blood cells, and platelets. Bounding boxes are generated around each detected cell as shown in figure 7.

The results show that:

- The model accurately detects multiple cells within a single image
- Overlapping cells are handled effectively
- Detection performance is consistent across different image conditions

8.2 WBC classification results



Figure 8: White blood cell classification output

Figure 8 shows the classification model categorizes detected white blood cells into their respective types based on morphological features.

Observations:

- The model accurately classifies major WBC types
- Feature extraction enables clear differentiation between cell types
- Classification results support further blast prediction

8.3 Blast prediction results

The blast prediction module identifies abnormal white blood cells and differentiates them from normal cells.

Observations:

- Blast cells are correctly identified based on learned features
- The system effectively distinguishes abnormal morphology
- This stage directly contributes to diagnostic interpretation

8.4 Detection performance evaluation

The performance of the blood cell detection model is evaluated using precision, recall, and mean average precision (mAP) as in Table 2.

Table 4 Performance metrics for blood cell detection

Class	**Precision**	**Recall**	**mAP@0.50**	**mAP@0.50:0.95**
WBC	0.989	0.996	0.994	0.907
RBC	0.978	0.942	0.990	0.921
Platelets	0.976	0.884	0.973	0.790
Overall	**0.981**	**0.941**	**0.986**	**0.872**

The detection model demonstrates strong performance across all blood cell classes. White blood cells show the highest detection accuracy, while platelet detection is comparatively lower due to their small size and variability.

8.5 WBC classification performance evaluation

The performance of the white blood cell classification model is evaluated using precision, recall, and F1-score as in Table 3.

Table 5 Performance metrics for WBC classification

WBC Class	**Precision**	**Recall**	**F1-Score**	**Support**
Basophil	0.99	1.00	1.00	349
Eosinophil	1.00	1.00	1.00	927
Lymphocyte	0.99	0.99	0.99	367
Monocyte	1.00	0.96	0.98	445
Neutrophil	0.98	1.00	0.99	1011
Overall Accuracy	-	-	**0.99**	3099
Macro Average	0.99	0.99	0.99	3099
Weighted Average	0.99	0.99	0.99	3099

The classification model achieves high performance across all classes, with near-perfect precision and recall values. The overall accuracy of 0.99 indicates reliable classification of white blood cells.

8.6 Discussion

The results demonstrate that the proposed system performs effectively across all stages of the pipeline, including detection, classification, and blast prediction. The combination of models contributes to improved accuracy and consistency in leukemia detection.

CHAPTER 9 DELIVERABLES

9.1 Software Deliverables

- Fully functional web-based diagnostic system developed using Streamlit
- Integrated deep learning pipeline:
  - YOLO-based cell detection (RBC, WBC, Platelets)
  - Vision Transformer (ViT) for WBC classification and blast detection
- Modular architecture including preprocessing, detection, inference, and reporting modules
  - Batch processing capability for multiple image analysis

9.2 Streamlit User Interface Deliverables

The system includes a comprehensive interactive UI designed for medical analysis and visualization and respective screenshots are in figures 9-13 .

Key UI Components

- Image upload interface (single & batch processing) - Figure 9
- Real-time detection visualization (original vs annotated images) - Figure 10
- Multi-tab dashboard (figure 11):
  - Batch Overview
  - Detection Results
  - WBC Analysis
- Combined Diagnostic Report -figure 12
- Final Blood Smear Diagnostic Report -figure 13
- Interactive charts (pie charts, bar graphs, heatmaps)
- Export options (CSV, ZIP reports)

UI Screenshots

1. Main Dashboard Interface

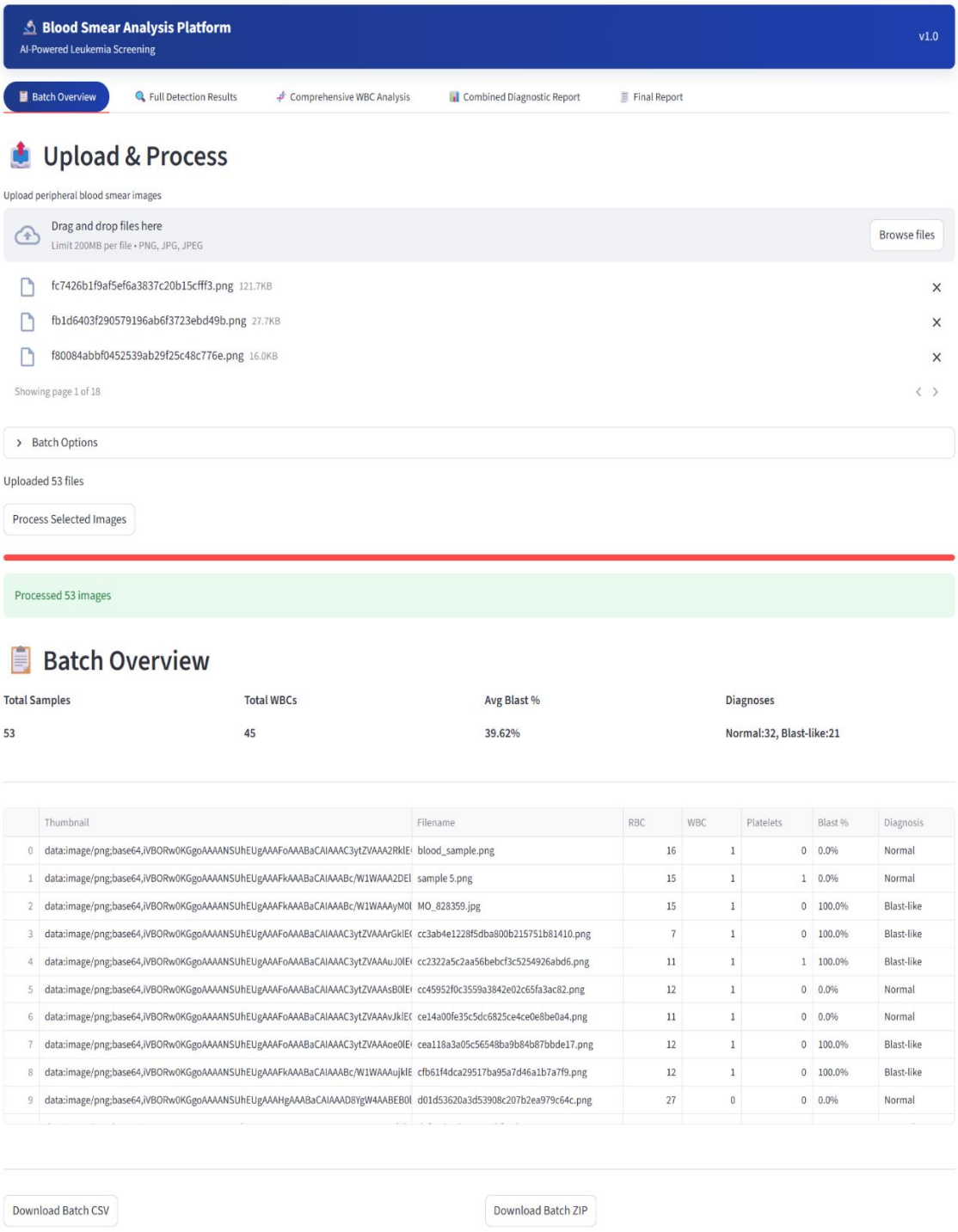


Figure 9: Streamlit Home / Upload Page

2. Detection Results View

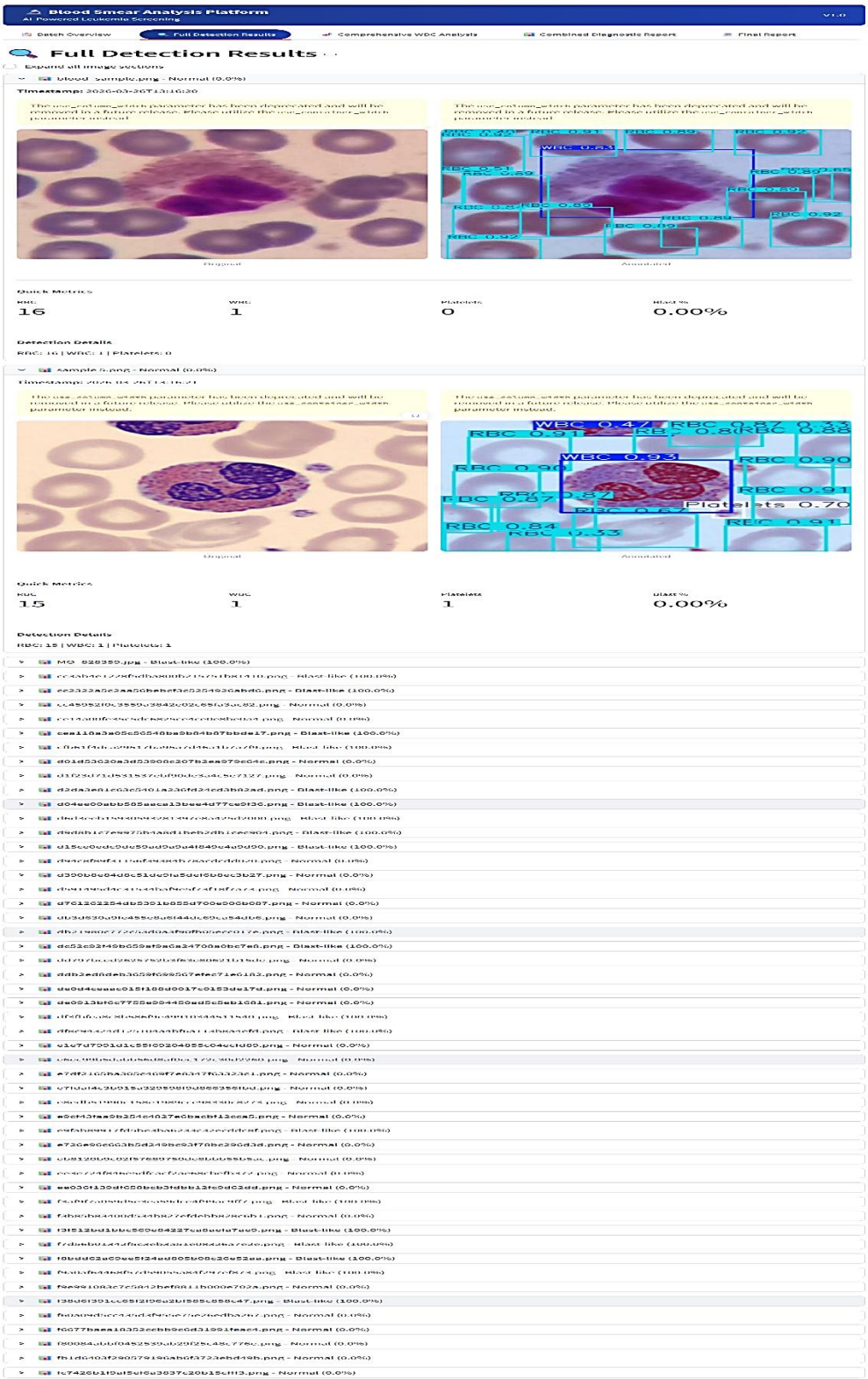


Figure 10 : Original vs Annotated Image Output

3. WBC Analysis Panel

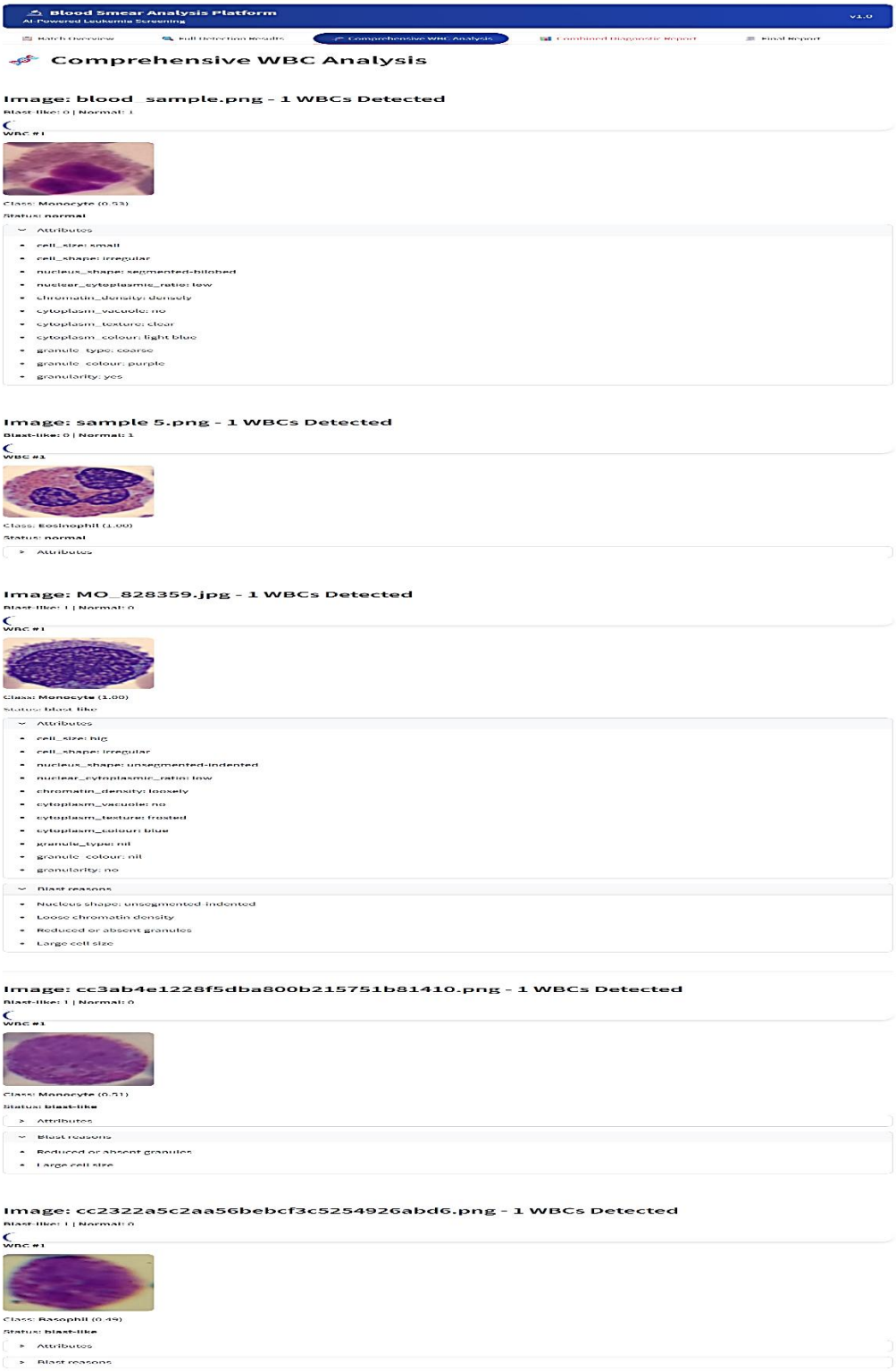


Figure 11: WBC Crops with Classification & Attributes

4. Combined Diagnostic Report

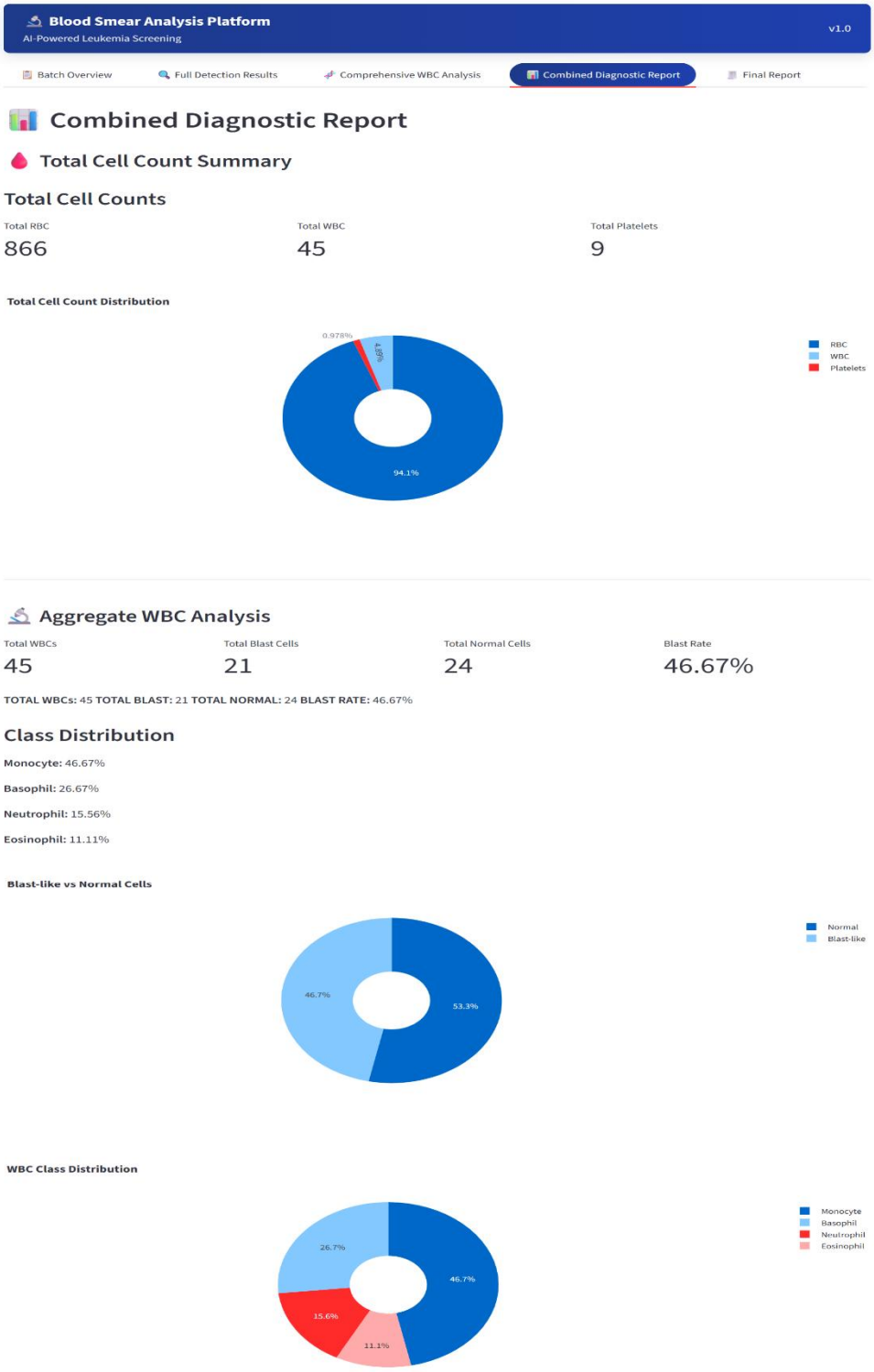


Figure 12: Charts & Aggregate Analysis

5. Final Diagnostic Report

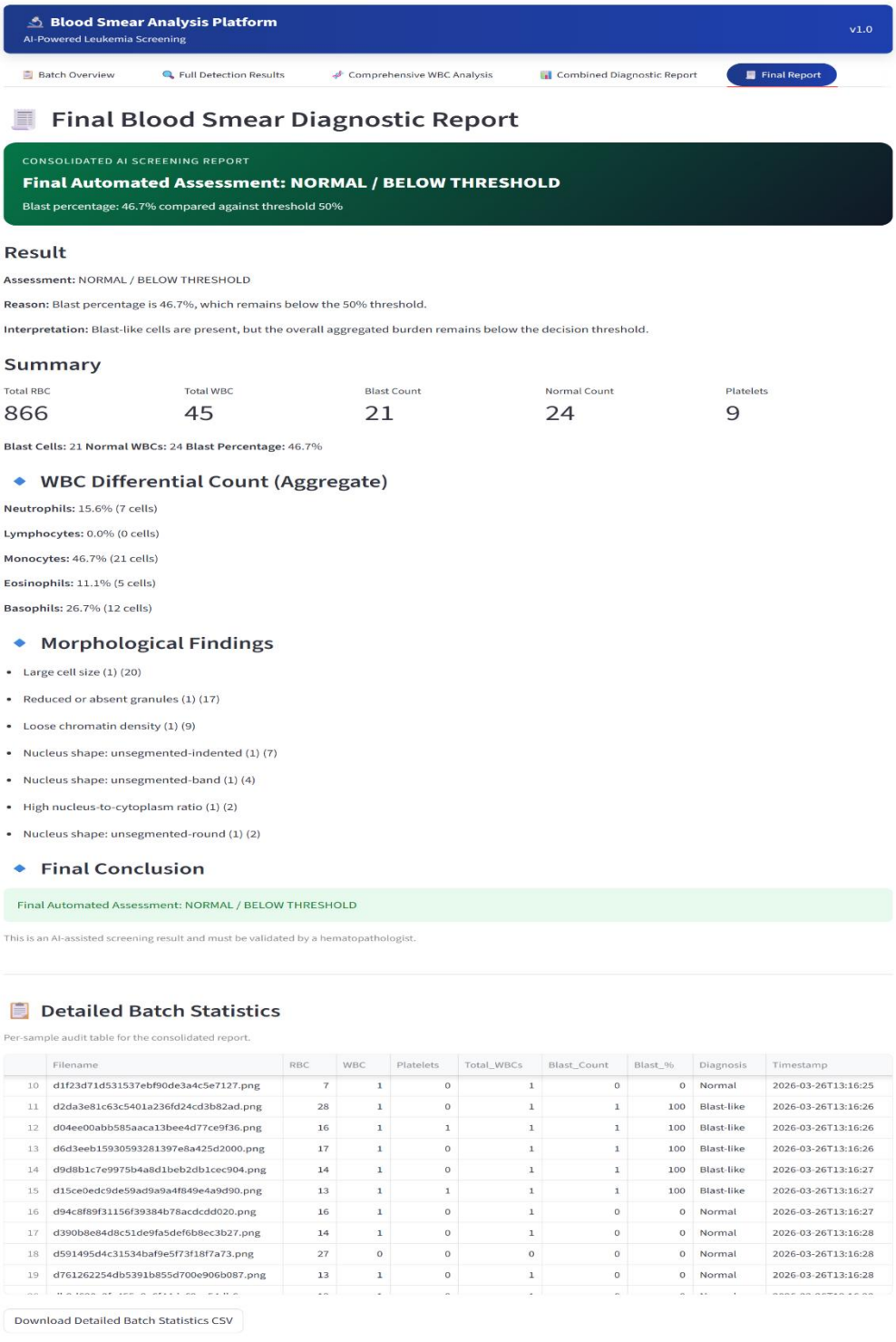


Figure 13: Final AI Decision with Blast % and Summary

9.3 Model Deliverables

- Trained YOLO model for blood cell detection
- Trained Vision Transformer model for:
    - WBC classification
    - Morphological feature extraction
    - Blast detection
- Model weights and checkpoint files
- Configurable thresholds (e.g., blast percentage threshold = 50%)

9.4 Dataset Deliverables

- Annotated blood smear image dataset
- YOLO-format label files
- Attribute-labeled dataset for WBC morphology
- Data preprocessing and augmentation pipeline

9.5 Diagnostic Output Deliverables

- Automated diagnostic report generation
- Quantitative outputs:
    - RBC count
    - WBC count
    - Platelet count
- Blast percentage calculation
- Classification results (Normal / Blast-like)
- Visual outputs with bounding boxes and labeled cells

9.6 Analytics and Visualization Deliverables

- Aggregate statistics across multiple samples
- WBC class distribution analysis

- Blast vs Normal comparison charts
- Heatmaps for multi-parameter analysis
- Comparative analysis across samples

9.7 Deployment Deliverables

- Deployable Streamlit application
- Requirements and dependency configuration
- Local deployment setup instructions
- Scalable structure for future cloud deployment

9.8 Documentation Deliverables

- Complete project report
- System architecture diagrams
- Workflow diagrams
- User manual for application usage
- Technical documentation for developers

9.9 Future-Ready Deliverables

- Extendable architecture for additional diseases
- Integration capability with hospital systems
- Support for real-time and large-scale batch analysis
- Scope for AI-assisted clinical decision support

9.10 Limitations and Future Work

While the proposed system demonstrates strong performance, several limitations are acknowledged:

9.10.1 Staining Protocol Dependency:

The system is trained on images acquired using specific staining protocols (primarily Giemsa and Leishman staining). Performance may vary when applied to images from laboratories using different staining methods without retraining or domain adaptation.

9.10.2 Diagnostic Threshold Simplification:

The blast percentage threshold of 50% used for leukemia indication is a simplification of clinical diagnostic criteria, which involve consideration of multiple parameters including patient history, bone marrow biopsy results, and cytogenetic analysis. The system is intended as decision support and must not replace formal clinical diagnosis.

9.10.3 Leukemia Subtype Classification:

The current system identifies the presence of blast cells but does not differentiate between leukemia subtypes such as Acute Lymphoblastic Leukemia (ALL), Acute Myeloid Leukemia (AML), and Chronic Lymphocytic Leukemia (CLL). Future work will extend the classification module to support subtype-level diagnosis.

9.10.4 Limited Clinical Validation:

The system has been evaluated on benchmark datasets under controlled conditions. Prospective clinical validation on patient samples from multiple institutions is required before deployment in a real clinical environment.

9.10.5 Future directions

The include integration with hospital Laboratory Information Systems (LIS), support for whole-slide image (WSI) analysis, extension to additional hematological diseases, and incorporation of explainability techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) to further enhance clinical trust.

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APPENDIX A:

This appendix provides supplementary data supporting the experimental results presented in Chapter 7 and Chapter 8. It includes detailed performance reports for both the detection and classification models, complete morphological attribute definitions, dataset composition details, the diagnostic decision rule summary, and the software environment configuration used during system development and evaluation.

A.1 Detailed WBC Classification Report

Table A.1 presents the complete per-class classification report generated from the test set evaluation of the Vision Transformer model on the WBC-ATT dataset. The report includes precision, recall, F1-score, and support values for each of the five white blood cell subtypes, along with overall accuracy and aggregate averages.

Table A.1: Detailed Per-Class Classification Report for WBC Subtypes

WBC Class	Precision	Recall	F1-Score	Support
Basophil	0.99	1.00	1.00	349
Eosinophil	1.00	1.00	1.00	927
Lymphocyte	0.99	0.99	0.99	367
Monocyte	1.00	0.96	0.98	445
Neutrophil	0.98	1.00	0.99	1011
Overall Accuracy	—	—	0.99	3099
Macro Average	0.99	0.99	0.99	3099
Weighted Average	0.99	0.99	0.99	3099

The classification model achieves near-perfect precision and recall across all five WBC subtypes. Eosinophil and Basophil classes achieve the highest individual F1-scores, while Monocyte shows a marginally lower recall of 0.96, attributable to morphological overlap with certain Lymphocyte and Neutrophil cell profiles.

A.2 Detailed Blood Cell Detection Report

Table A.2 presents the complete per-class detection performance of the YOLOv8-based model evaluated on the TXL-PBC test set. Performance is reported using precision, recall, mean average precision at IoU threshold 0.50 (mAP@0.50), and the stricter mAP@0.50:0.95 metric.

Table A.2: Detailed Per-Class Detection Performance on TXL-PBC Test Set

Class	Precision	Recall	mAP@0.50	mAP@0.50:0.95
WBC	0.989	0.996	0.994	0.907
RBC	0.978	0.942	0.990	0.921
Platelets	0.976	0.884	0.973	0.790
Overall	0.981	0.941	0.986	0.872

White blood cells achieve the highest detection accuracy across all metrics, reflecting their distinctive morphological appearance and relatively larger size within blood smear images. Platelet detection shows the lowest mAP@0.50:0.95 of 0.790, which is expected given the small size and irregular clustering patterns of platelets in peripheral blood smear images.

A.3 Morphological Attribute Definitions and Class Labels

Table A.3 lists all eleven morphological attributes used in the multitask Vision Transformer classification framework. Each attribute corresponds to an independent classification head in the multitask architecture, and together they characterize the complete morphological profile of each detected white blood cell. The WBC-ATT dataset provides over 113,000 attribute-level annotations across these eleven categories.

Table A.3: Morphological Attribute Definitions and Class Labels

Attribute	Number of Classes	Class Labels
Cell Size	2	Big, Small
Cell Shape	2	Irregular, Round
Nucleus Shape	6	Irregular, Segmented-Bilobed, Segmented-Multilobed, Unsegmented-Band, Unsegmented-Indented, Unsegmented-Round
Nuclear-to-Cytoplasmic Ratio	2	High, Low
Chromatin Density	2	Densely Packed, Loosely Packed
Cytoplasm Vacuoles	2	Present, Absent
Cytoplasm Texture	2	Clear, Frosted
Cytoplasm Color	3	Blue, Light Blue, Purple Blue
Granule Type	4	Coarse, Absent, Round, Small
Granule Color	4	Absent, Pink, Purple, Red
Granularity	2	Present, Absent
Total Attribute Labels	—	113,000+

These eleven attributes collectively enable interpretable and clinically grounded blast cell identification, distinguishing the proposed system from approaches that rely solely on binary blast or normal labels.

A.4 WBC Subtype Distribution in WBC-ATT Dataset

Table A.4 presents the distribution of white blood cell subtypes within the complete WBC-ATT dataset. Stratified splitting was applied during dataset partitioning to ensure proportional class representation across training, validation, and test subsets.

Table A.4: WBC Subtype Distribution in the WBC-ATT Dataset

WBC Subtype	Total Images	Percentage
Neutrophil	3,329	32.3%

Eosinophil	3,117	30.3%
Monocyte	1,420	13.8%
Basophil	1,218	11.8%
Lymphocyte	1,214	11.8%
Total	10,298	100%

Neutrophils and Eosinophils constitute the majority of the dataset, collectively accounting for approximately 62.6% of all images. Monocyte, Basophil, and Lymphocyte classes are moderately underrepresented relative to their clinical frequency, which is addressed through stratified sampling and data augmentation during training.

A.5 Source Dataset Composition of TXL-PBC

Table A.5 details the individual source datasets that were aggregated to form the TXL-PBC dataset used for blood cell detection model training. The TXL-PBC dataset was constructed to address the limitations of individual public datasets, including limited size, inconsistent annotations, and narrow coverage of cell types.

Table A.5: Source Dataset Composition of TXL-PBC

Source Dataset	Description	Approximate Size
BCCD	Annotated blood smear images for RBC, WBC, and platelet detection	~12,500 images
BCDD	Peripheral blood smear dataset for blood cell localization	Multiple annotated images
PBC	Labeled peripheral blood smear images for WBC subtype classification	Multiple labeled images
Raabin-WBC	Large-scale dataset with detailed WBC subtype annotations	~17,000 images
Combined TXL-PBC	High-quality microscopic images with unified annotations	1,260 images / 18,143 bounding boxes

The combined TXL-PBC dataset provides 18,143 bounding box annotations covering RBCs, WBCs, and platelets across 1,260 high-quality microscopic images, offering a diverse and representative training corpus for the detection model.

A.6 Diagnostic Decision Rule Summary

Table A.6 summarizes the threshold-based diagnostic decision rule applied by the system following blast percentage computation. The blast percentage is calculated as the ratio of detected blast cells to the total white blood cell count within the analyzed image.

Table A.6: Diagnostic Decision Rule Based on Blast Percentage

Blast Percentage	Diagnostic Interpretation	Clinical Implication
≥ 50%	Leukemia Likely	High concentration of blast cells detected; immediate clinical review recommended
20% – 49%	Borderline / Inconclusive	Elevated blast count observed; further hematological investigation advised
< 20%	Non-Leukemic	Blast cell concentration within acceptable range; no immediate concern indicated

Note: The 50% threshold applied in this system represents a computational decision boundary designed for automated screening purposes. It is aligned with established clinical criteria for acute leukemia diagnosis, where a blast percentage of 20% or above in bone marrow or peripheral blood is a recognized diagnostic indicator. Clinical confirmation by a qualified hematologist, supported by bone marrow biopsy and cytogenetic analysis, is required before any definitive diagnosis is made.

A.7 Software Dependencies and Version Configuration

Table A.7 lists all key software libraries and frameworks used during the development, training, evaluation, and deployment of the proposed leukemia detection system. All components are open-source and compatible with standard GPU-accelerated computing environments.

Table A.7: Software Dependencies and Versions

Software / Library	Version	Purpose
Python	3.9+	Core programming language
PyTorch	2.0+	Deep learning framework for model training and inference
Ultralytics YOLOv8	8.0+	Blood cell detection model implementation
Timm (PyTorch Image Models)	0.9+	Vision Transformer architecture implementation
OpenCV	4.7+	Image preprocessing, annotation, and visualization
Streamlit	1.25+	Web-based diagnostic user interface
NumPy	1.24+	Numerical computation and array operations
Pandas	2.0+	Data handling, tabulation, and report generation
Matplotlib / Seaborn	3.7+	Visualization, charting, and heatmap generation
Scikit-learn	1.3+	Evaluation metrics computation and classification reports
CUDA	11.8+	GPU acceleration for model training and inference

GIT HUB LINK

LINK: HTTPS://GITHUB.COM/MADHU0096/CAPSTONE-PROJECT-LEUKEMIA-DETECTION-FROM-BLOOD-SMEAR-IMAGES